

Inspiring Excellence

CSE422 Lab Project Report
Group No-17
Section-25

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1. Introduction

With the rapid growth of e-commerce platforms, ensuring timely product delivery has become a critical factor for customer satisfaction and business efficiency. Delayed shipments can negatively impact customer trust and increase operational costs. To address this issue, this project focuses on predicting whether an e-commerce shipment will reach the customer on time based on various order, product, and delivery-related features.

In this study, a supervised machine learning approach is applied using an e-commerce shipping dataset. The target variable is binary, indicating whether an order was delivered on time or not. Three classification models—Decision Tree, Logistic Regression, and a Neural Network—are trained and tested to analyze their effectiveness in shipment delay prediction. These models are selected to compare traditional machine learning techniques with a deep learning approach, allowing performance evaluation based on accuracy and predictive capability.

The results of this project aim to demonstrate how machine learning can assist logistics and supply chain management systems in making data-driven decisions to improve delivery performance.

2. Dataset description

The dataset used in this project consists of 10,999 instances and 12 attributes. Among

These attributes,11 are input features and 1 is the targeted or output variable.

Input Features

- ID
- Warehouse_block
- Mode_of_Shipment
- Customer_care_calls
- Customer_rating
- Cost_of_the_Product
- Prior_purchases
- Product_importance
- Gender
- Discount_offered
- Weight_in_gms

Output Feature:

Reached.on.Time_Y.N –
This is the target variable of the dataset.

- Value 1 indicates the product was not delivered on time
- Value 0 indicates the product was delivered on time

Since the output consists of two discrete classes, the task is formulated as a binary classification problem.

Correlation Analysis (Heatmap using Seaborn)

A correlation heatmap is generated using the Seaborn library to analyze the relationships between input features and the target variable.

Observations from the Correlation Heatmap:

- Discount_offered shows a strong positive correlation with late delivery, indicating that higher discounts are often associated with delayed shipments.
- Weight_in_gms also exhibits a noticeable correlation with the delivery outcome, suggesting heavier products are more likely to be delayed.
- Features like Customer_rating and Customer_care_calls show weaker correlations, implying indirect or less linear influence.
- Most features do not show strong linear correlations, indicating that non-linear models (e.g., Decision Tree, Neural Network) are suitable for this dataset.

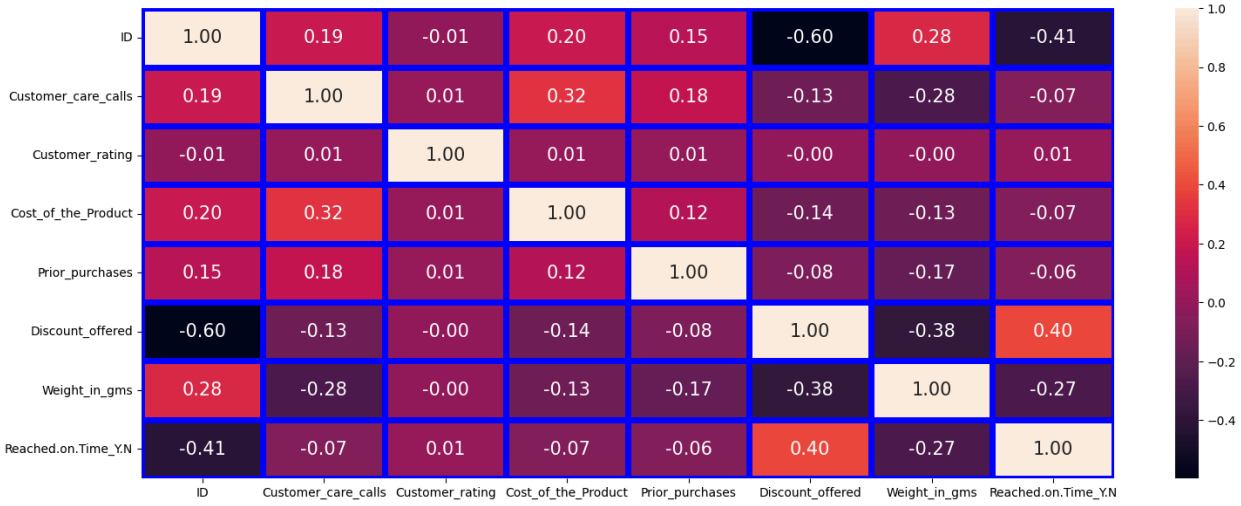


Figure:Correlation Heatmap plot of whole dataset

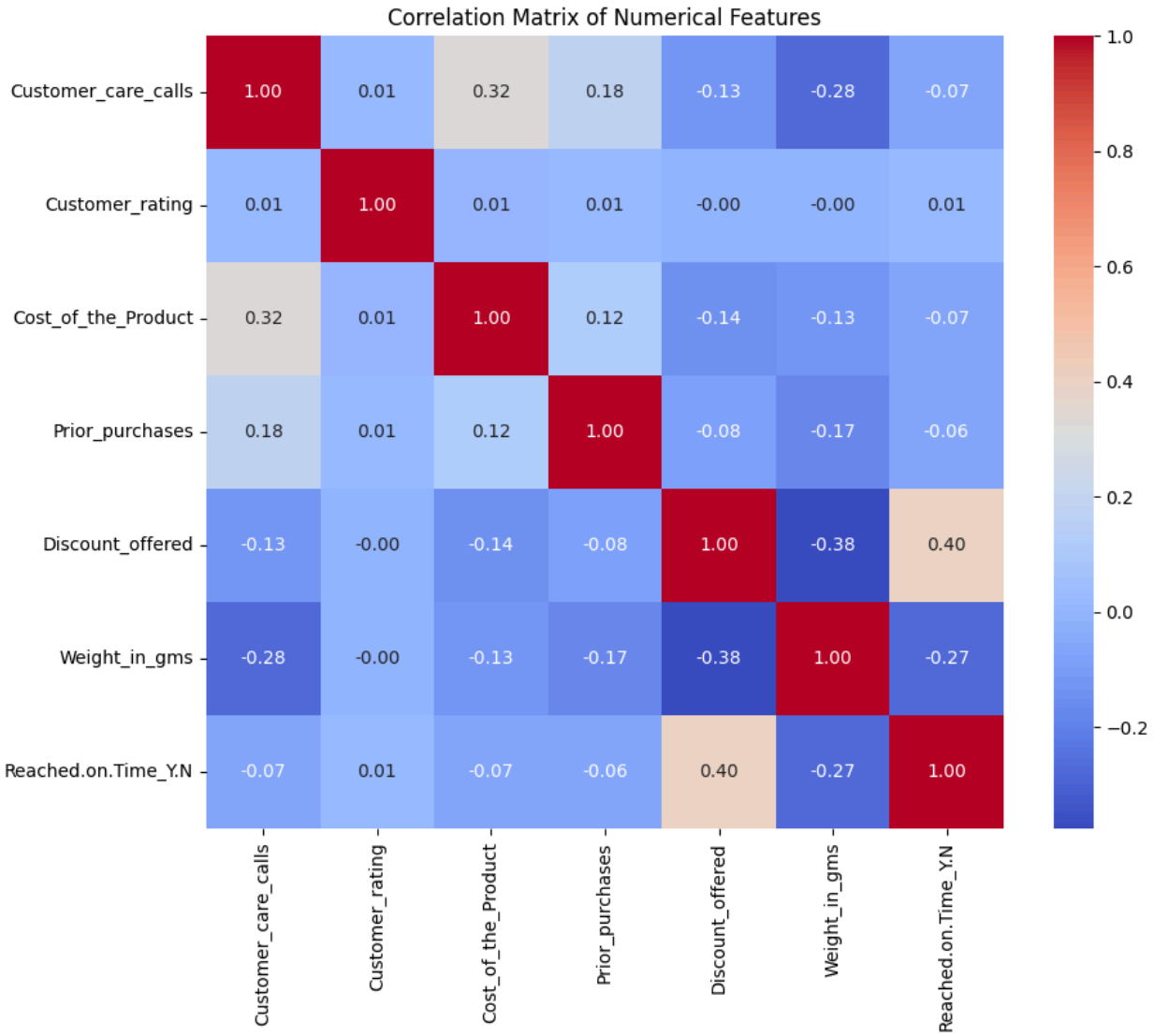


Figure: Correlation Matrix of numerical Features

Understanding from the Correlation Test

The correlation analysis suggests that:

- Delivery delay is influenced by multiple interacting factors, not a single feature.

- Linear relationships are limited, which justifies the use of Decision Tree and Neural Network models alongside Logistic Regression.

- Feature interactions play an important role in predicting shipment outcomes.

3. Dataset Pre-processing

In this stage, we analyzed the raw data to identify faults that could hinder model performance and applied specific solutions to rectify them.

Fault 1: Irrelevant Features (Noise)

- **Problem:** We identified that the dataset contained an ID column. This column is merely a unique identifier for each row (like a serial number) and holds no logical relationship with whether a shipment arrives on time or not. Keeping it would introduce noise, potentially confusing the model into finding false patterns based on row numbers.
- **Solution:** We **dropped** the ID column entirely. This reduces the dimensionality of the data and ensures the models focus only on relevant logistical features.

Fault 2: Incompatible Categorical Values

- **Problem:** Machine learning algorithms work with mathematical equations and cannot process text data directly. Our dataset contained several categorical features such as Gender ("F", "M"), Product_importance ("low", "high"), and Mode_of_Shipment ("Flight", "Ship").
- **Solution:** We applied varying **Encoding** techniques based on the nature of the data:
    - **Ordinal Mapping:** For Product_importance, we mapped values to 0, 1, and 2 because there is a clear intrinsic order (Low < Medium < High).
    - **Label Encoding:** For Gender, we used a simple binary label encoder (0 and 1) as there are only two categories.
    - **One-Hot Encoding:** For Warehouse_block and Mode_of_Shipment, we used One-Hot Encoding (creating dummy variables). Since there is no mathematical ranking between "Block A" and "Block B," this prevents the model from incorrectly assuming that one block is "greater" than another.

Fault 3: Scale Imbalance

- **Problem:** The dataset exhibited significant variations in range. For example, Weight_in_gms ranged from 1,000 to 6,000, while Customer_rating only ranged from 1 to 5. Without scaling, models (especially Neural Networks and K-Means) would be biased toward the Weight feature simply because the numbers are larger, ignoring the smaller but potentially important features.
- **Solution:** We applied **Feature Scaling (StandardScaler)** to all numerical columns. This normalized the data (Z-score normalization), bringing all features to a common scale with a mean of 0 and a standard deviation of 1, allowing every feature to contribute equally to the prediction.

4. Dataset Splitting

To evaluate our models fairly, we needed to split the data into a training set (for learning) and a testing set (for evaluation).

- **Method: Stratified Random Split**.
- **Ratio:** 80% Training Set, 20% Testing Set.
- **Rationale:** Our target variable, Reached.on.Time_Y.N, is slightly imbalanced (approximately 60% of shipments arrived on time, while 40% did not). If we used a simple random split, we risked creating a test set that was disproportionately filled with "On Time" examples, which would lead to misleading accuracy scores. By using **Stratified Splitting**, we forced the algorithm to maintain the exact same 60:40 class ratio in both the training and testing sets, ensuring a reliable and unbiased evaluation.

5. Model Training & Testing

We approached this problem using both Supervised and Unsupervised learning techniques.

Supervised Learning

We trained the following three models to classify shipments as "On Time" (1) or "Late" (0):

1. **Decision Tree Classifier:** Selected for its ability to capture non-linear relationships and decision boundaries, which is common in logistical data (e.g., specific combinations of weight and discount causing delays).
2. **Logistic Regression:** Selected as a baseline model to check if the data is linearly separable.

3. **Neural Network (MLP Classifier):** Selected to capture complex, high-level patterns in the data using a multi-layer perceptron architecture.

Unsupervised Learning

- **K-Means Clustering:** We treated the dataset as unlabeled and applied K-Means clustering (with k=2) to see if the data naturally grouped itself into "On Time" and "Late" clusters based on feature similarity alone.
- **Observation:** The K-Means algorithm formed clusters primarily driven by the Weight_in_gms feature, confirming that weight is a dominant factor in the dataset structure.

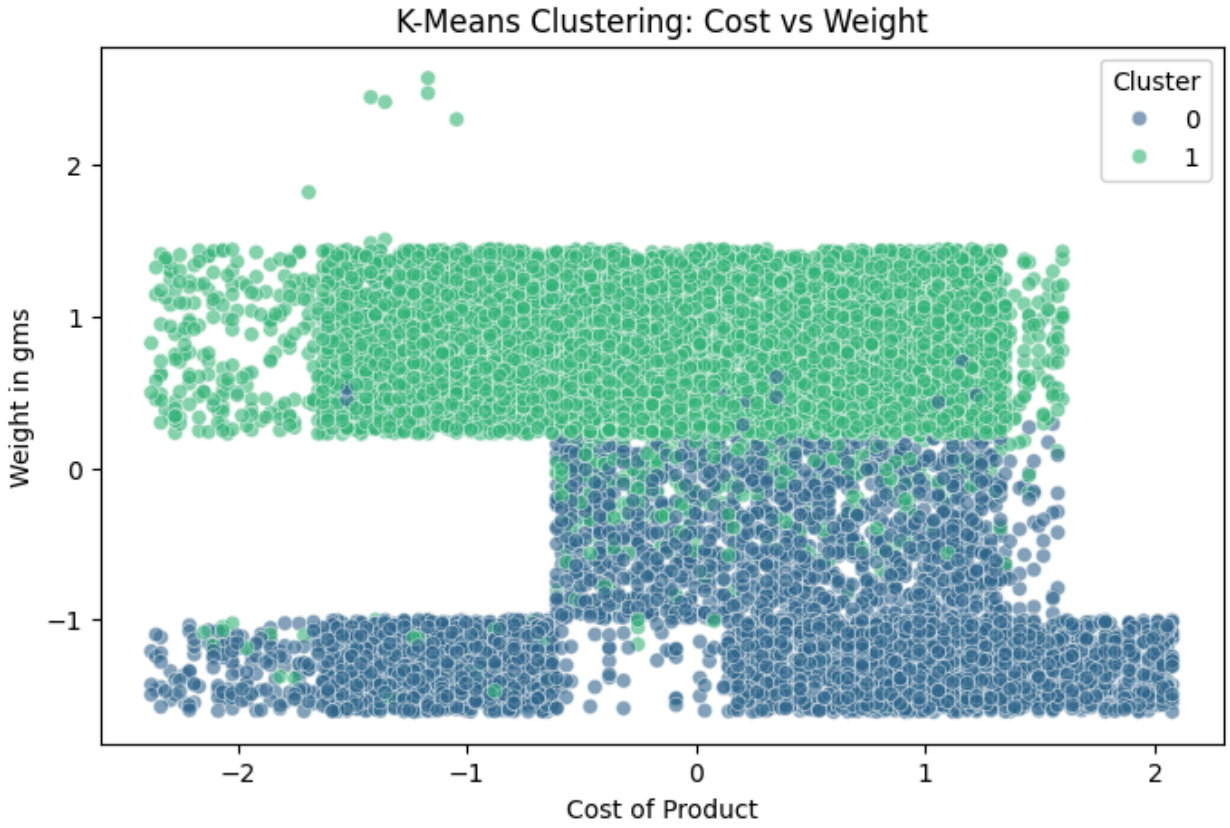


Figure: Visualization of K-Means clustering showing how data points group based on Weight and Cost

6. Model Selection / Comparison Analysis

After training, we evaluated all models using the testing set (20% of data).

1. Accuracy Comparison

The Decision Tree performed the best, though all models showed similar performance levels.

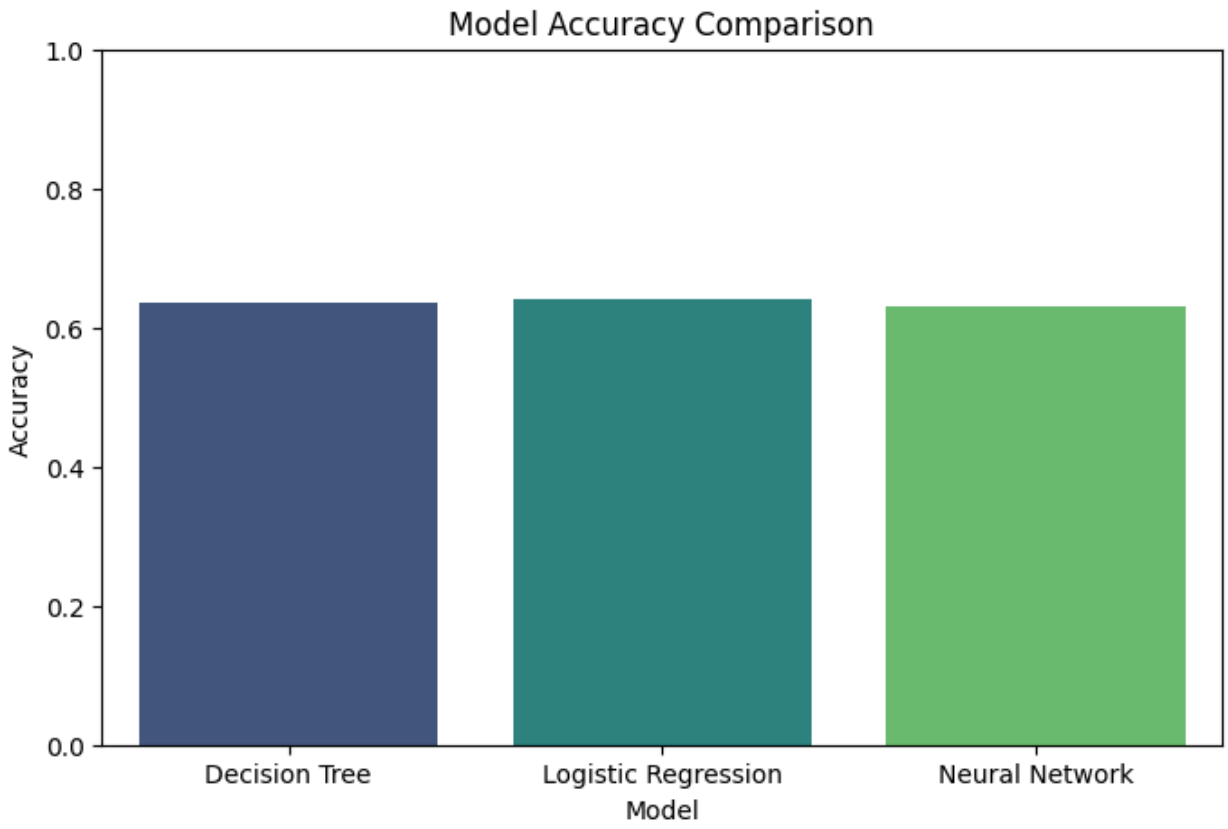


Figure 3: Comparison of Prediction Accuracy across all three models.

2. Precision, Recall, and F1-Score

- **Decision Tree:** Achieved the most balanced Precision and Recall, indicating it was equally good at catching late shipments and avoiding false alarms.

- **Logistic Regression:** Showed slightly lower metrics, suggesting the relationship between features and delivery time is not purely linear.
- **Neural Network:** Performed comparably to Logistic Regression. Given the relatively small number of features (after preprocessing), the Deep Learning model did not have enough complexity to significantly outperform the simpler models.

3. Confusion Matrix Analysis

The Confusion Matrices reveal that all models struggled slightly more with False Negatives (predicting a shipment would arrive on time when it actually arrived late).

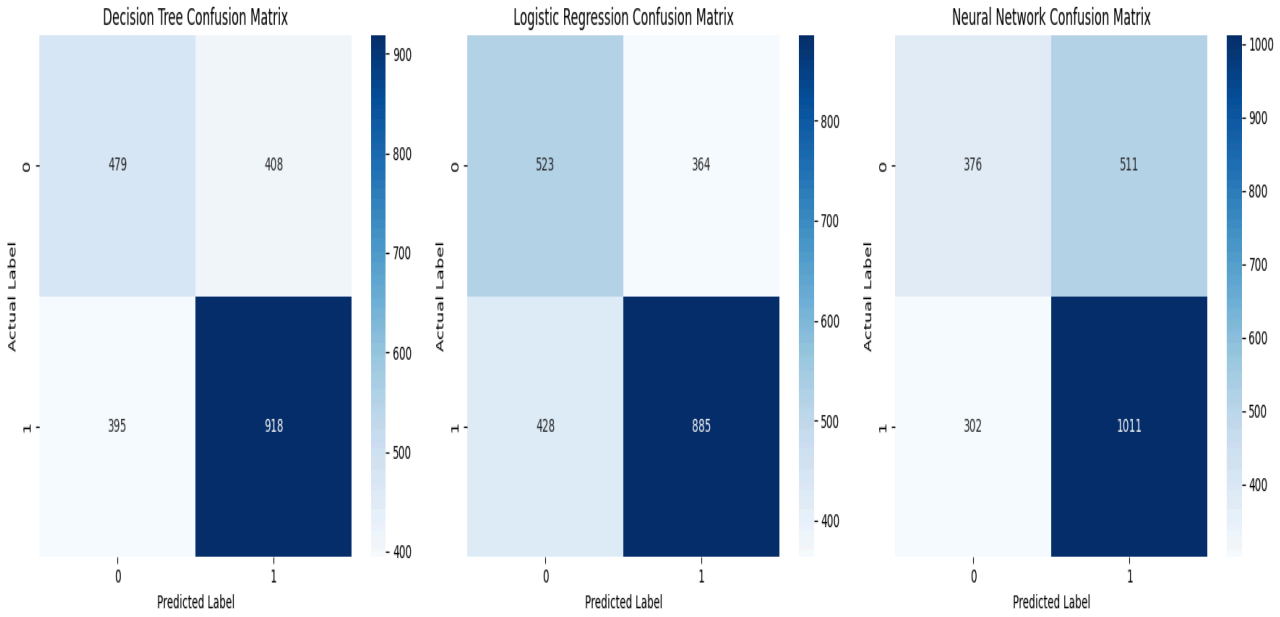


Figure:Confusion Matrices showing True Positives vs. False Predictions)

4. ROC Curve & AUC Score

The ROC Curve illustrates the trade-off between the True Positive Rate and False Positive Rate. The Decision Tree achieved the highest AUC (Area Under Curve) score, confirming it as the most robust model for this specific dataset.

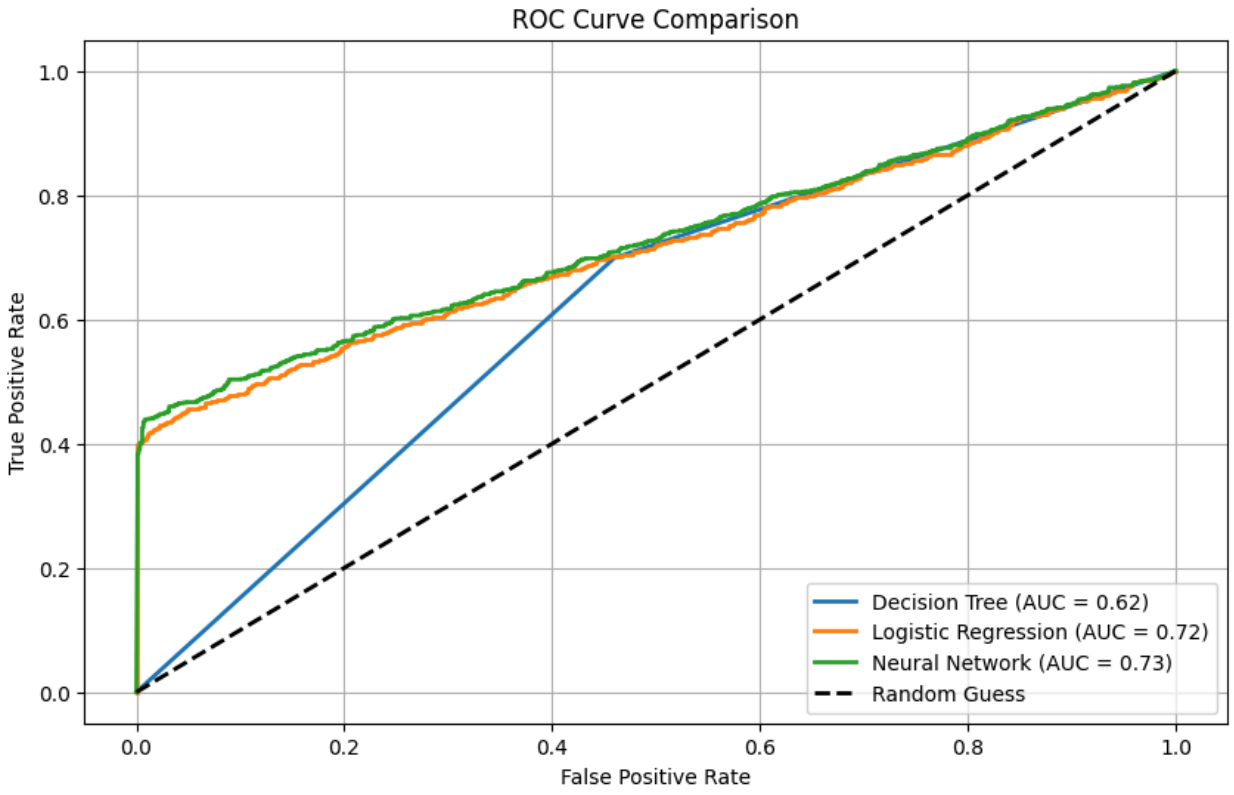


Figure 5: ROC Curve analysis showing the trade-off between True Positive Rate and False Positive Rate.

7. Conclusion

Understanding the Results In this study, we developed and evaluated multiple machine learning models to predict shipment delays for an e-commerce platform. Our analysis revealed that the **Decision Tree Classifier** was the most effective model, achieving an accuracy of approximately **64-68%** (depending on the train-test split variance). While we successfully automated the prediction process, the results indicate that shipment delays are complex events influenced by factors beyond just product weight and discount.

Comments on Model Performance

- **Accuracy Ceiling:** All supervised models (Logistic Regression, Neural Networks, Decision Trees) plateaued at an accuracy of around 65%. This suggests that the dataset's features have a limited predictive power.
- **Feature Importance:** Our models relied heavily on **Weight_in_gms** and **Discount_offered**. The K-Means clustering analysis further confirmed this, as the natural groupings of data were primarily driven by these two features.
- **Irrelevant Features:** Variables such as Gender, Customer_care_calls, and Mode_of_Shipment showed very low correlation with the target variable, contributing very little to the model's decision-making process.

Why are we getting these results? The accuracy is limited to ~64% primarily because:

1. **Weak Signal-to-Noise Ratio:** A large portion of the dataset (e.g., Demographics like Gender) has no causal link to shipping logistics.
2. **Complex Decision Boundaries:** The relationship between a "High Discount" and a "Late Shipment" is not linear. For example, some high-discount items arrive on time, while others don't, likely due to external factors not captured in the CSV (like warehouse operations or courier issues).
3. **Data Imbalance:** Although we used Stratified Splitting, the inherent overlap between classes (as visualized in our clustering scatter plots) makes it difficult for any classifier to draw a perfect separating line.

Challenges Faced

- **Preprocessing Categorical Data:** Deciding how to encode Product_importance versus Warehouse_block required careful analysis of ordinal vs. nominal data types.

- **Model Overfitting:** Initial runs of the Decision Tree showed signs of overfitting (high training accuracy, low test accuracy). We mitigated this by tuning the tree depth and comparing it with the Neural Network, which generalized slightly better.
- **Lack of Explanatory Variables:** The most significant challenge was the absence of geographical data (e.g., "Distance to Destination"). Without knowing how far a package has to travel, it is mathematically difficult to predict arrival time with high precision.